

Compiler Design

CA-1

F.M.-20

Time – 1Hr

Date – 24/03/2025

1. Write RE for all strings of digits that represent even numbers. [2]

2. Consider the following C program segment: [5]

```
int main() {  
    int a=5, a3c;  
    / this code to find lexemes  
    printf("Hello World!");  
    Return (0);  
}
```

- List the lexemes and corresponding tokens.
- List the errors detected in lexical analysis phase.

3. Consider the following RE: [8]

$$(a|c)^*b(a|c)^*$$

- Draw syntax tree.
- Compute *nullable()*, *firstpos()*, *lastpos()* and *followpos()*.
- Construct DFA using (a) and (b).

4. Convert the following RE into an NFA by Ken Thompson method.

$$(a|c)^*b(a|c)^*$$

[5]



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1. Write RE for all strings of  $a$ 's and  $b$ 's that contain an even number of  $a$ 's and an even number of  $b$ 's. [2]
2. Consider the following C program segment: [5]  

```
int add() {  
    int a=5, b=3, ac=4;  
    /* this code to find lexemes */  
    printf "Hello World!";  
    return (a+b);  
}
```

  - a. List the lexemes and corresponding tokens.
  - b. List the errors detected in lexical analysis phase.
3. Consider the following RE: [8]  
 $(aa|b)^*ca^*$ 
  - a. Draw syntax tree.
  - b. Compute  $nullable()$ ,  $firstpos()$ ,  $lastpos()$  and  $followpos()$ .
  - c. Construct DFA using (a) and (b).
4. Convert the RE of Q. No. - 3 into an NFA by Ken Thompson method. [5]



1 - null

First  
e1 - null then e → Union  
the f(e1)

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1. Write RE for all strings of  $a$ 's,  $b$ 's and  $c$ 's in which every  $a$  is immediately followed by at least two  $b$ 's. [2]
2. Consider the following C program segment: [5]  

```
Int S=a, a3c,  
/// this code to find lexemes  
int x = add(a, a3c);  
printf("Sum is = %d",x);
```

  - a. List the lexemes and corresponding tokens.
  - b. List the errors, if detected in lexical analysis phase.
3. Consider the following RE: [8]  
 $(a|b)^*a(b|a)^*$ 
  - a. Draw syntax tree.
  - b. Compute *nullable()*, *firstpos()*, *lastpos()* and *followpos()*.
  - c. Construct DFA using (a) and (b).
4. Convert the RE of Q. No. – 3 into an NFA by Ken Thompson method. [5]



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1. Write RE for all strings of  $a$ 's and  $b$ 's that start with  $a$  and have odd length. [2]

2. Consider the following C program segment: [5]

```
while (I > 10){  
    i = I - 1;  
    / ***** this code to find lexemes */  
    printf("Inside loop!!");  
    int c = a + b;  
}
```

- List the lexemes and corresponding tokens.
- List the errors, if detected in lexical analysis phase.

3. Consider the following RE: [8]

$(aa)^*c(ab)^*$

- Draw syntax tree.
- Compute *nullable()*, *firstpos()*, *lastpos()* and *followpos()*.
- Construct DFA using (a) and (b).

4. Convert the RE of Q. No. – 3 into an NFA by Ken Thompson method. [5]

